



MSc Data Science

Machine Learning for CO₂ Storage Site Suitability: A Data-Driven Approach Using Geospatial Analysis

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Abstract

This study employs Firth-penalised logistic regression to assess the suitability of geological formations for carbon dioxide, with a small sample size ($n = 56$; 35 suitable, 21 unsuitable) drawn from the North Sea. Four petrophysical parameters were investigated: porosity, depth, permeability, and seal thickness (*crThickness*). Porosity emerged as the strongest positive predictor (OR = 1.63; 95% CI: 0.72–3.68), indicating that more porous formations are associated with a greater probability of suitability, in accordance with established reservoir quality principles. Conversely, seal thickness (OR = 0.68; 95% CI: 0.34–1.35) and permeability (OR = 0.82; 95% CI: 0.38–1.76) exhibited negative associations, whilst depth demonstrated a negligible effect (OR = 1.01; 95% CI: 0.57–1.80). Nevertheless, none of the predictors attained statistical significance, as all 95% confidence intervals included unity, reflecting the constraints of the available dataset. The model achieved a cross-validated accuracy of 54% and a Brier Score of 0.259, suggesting performance marginally above chance with limited generalisation to unseen data. The low Pseudo R^2 (0.052) further corroborates that the selected parameters alone are insufficient to reliably classify formation suitability. In spite of these limitations, Firth-penalised regression demonstrates considerable promise for small, imbalanced geological datasets, offering a statistically principled framework for early-stage CCS site screening. Future work should incorporate additional geological and geomechanical parameters to improve model performance, providing early-stage screening insights to support CCS deployment in alignment with Shell's 2050 net-zero commitments.

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1 Introduction

In response to the significant environmental impacts of carbon dioxide (CO₂) emissions from industrial sources such as fuel combustion in boilers, furnaces, and gas turbines, as well as flaring and venting during refining and hydrocarbon production, increasing attention is being directed towards mitigation strategies (United States Energy Association 2023, American Institute of Chemical Engineers 2024, ClientEarth 2024). Carbon Capture and Storage (CCS) is one of the emerging approaches. It involves capturing CO₂ from industrial sources and securely storing it within subsurface geological formations. Shell, as a leading global energy company, has committed to achieving net-zero emissions by 2050, with CCS forming a central element of its strategy (Shell plc 2025). Although CCS offers considerable potential, its effectiveness depends on the identification of suitable storage sites. Applying data science to analyse geological datasets, model subsurface storage capacity, and assess leakage risks can enhance the scalability, safety, and efficiency of CCS projects. This reflects a transition from enhanced oil recovery (EOR) towards contemporary environmental objectives (Bajpai et al. 2022, Guo et al. 2021, Nassabeh et al. 2023b).

1.1 Research Questions

1. How can data science be applied to identify high-potential geological storage sites for CO₂?
2. What machine learning models are most effective for predicting site suitability?
3. How can probabilistic modelling aid in understanding containment risks?
4. What visualisation techniques can enhance stakeholder insight into site selection?

2 Background Literature

2.1 Relevant Literature

The first ever injection of CO₂ in geological formations started in 1996 a distinguished project called the Sleipner. It is the first commercial-scale implementation of carbon dioxide storage within deep saline aquifers globally. In 1998, a complementary initiative known as the Saline Aquifer CO₂ Storage (SACS) project was established to monitor the injected carbon dioxide (Torp & Gale 2004). Global efforts have increasingly focused on evaluating the reliability and effectiveness of carbon capture, utilisation and storage (CCUS) within a variety of subsurface environments (Wright et al. 2009, Eiken et al. 2011). The extensive experience of the petroleum industry in applying CO₂ injection for enhanced oil recovery has played a pivotal role in refining methods of geological sequestration, thereby encouraging renewed consideration of different formations as potential repositories for long-term storage (Harrison & Falcone 2014, Farajzadeh et al. 2020). Recent research has underscored the complexity of selecting suitable storage sites, highlighting the importance of rigorous technical frameworks underpinned by comprehensive data (Nassabeh et al. 2023a, Wright et al. 2009). Effective assessment depends upon the integration of geological, environmental and oceanographic information to guarantee not only security and functionality but also economic practicality. Among the principal candidates for offshore storage are depleted hydrocarbon reservoirs, deep saline systems and coal seams that are unsuitable for mining, each of which must be subjected to stringent evaluation frameworks and ranking methodologies based on geological complexity (Eigbe et al. 2023, Yao et al. 2023).

Traditional approaches to screening offshore CO₂ storage sites face a number of practical challenges. One concern is the possibility of leakage from geological formations, highlighting the value of careful assessment of subsurface conditions at candidate sites. Regulatory requirements and public perceptions of CO₂ storage can also add complexity to the process. Data-driven models and machine learning (ML) techniques offer an alternative that can bring additional flexibility and accuracy to site evaluations. These approaches can combine a wide range of geological, environmental, and operational information to estimate site suitability and potential risks. ML techniques are particularly useful in recognising subtle patterns and relationships in the data, which can help build a more detailed understanding of geological characteristics and leakage risks. At the same time, there are considerations when using these tools, such as the need for sufficiently large and varied datasets, the expertise required to develop and apply the models, and the possibility that data biases could affect predictions. In addition, the way ML results are interpreted may require careful attention, especially in regulatory and public engagement contexts (Nassabeh et al. 2024, Vo Thanh et al. 2023, Naghizadeh et al. 2024). Integrating the accuracy of conventional site screening with the predictive capabilities of advanced analytical methods

can improve the effectiveness of identifying suitable geological CO₂ storage locations, while addressing the shortcomings of each method (Wallmann et al. 2015, Araujo de Itriago & Swearingen 2011). Sustainable energy development depends on innovative approaches, where tools like seismic imaging and machine learning play a vital role in pinpointing suitable sites for the effective implementation of CCUS projects (Nassabeh et al. 2024). Evaluating the suitability of geological settings is essential, requiring investigations to identify subsurface formations, such as saline aquifers or depleted petroleum reservoirs, exhibiting the necessary porosity and permeability (Aminu et al. 2017). Thorough assessment of potential sites is crucial, taking into account parameters such as reservoir depth, existing fault structures, and the characteristics of the cap rock (Dai et al. 2014).

Current research has highlighted the importance of adopting a comprehensive approach to the selection of CO₂ storage sites, taking into account geological conditions, economic viability, and risk management aspects. Candidate formations are typically assessed against screening parameters such as reservoir depth, brine salinity, and the likelihood of obstructing existing industrial activities, while ranking frameworks are employed to address geological complexity. Such evaluations apply rigorous technical standards to guarantee the long-term security of storage and to prevent adverse impacts on human health, groundwater resources, climate stability, environmental quality, and in particular, marine ecosystems. Ku et al. (2023) provide a comprehensive review of progress in offshore carbon storage, offering insights into both the present state and the anticipated developments of carbon capture and storage (CCS) (Ku et al. 2023). Their analysis emphasises the difficulties and ambiguities linked to marine ecosystem-based carbon dioxide removal (CDR). The study further discusses the processes, potential sites, advantages, and environmental risks associated with injecting CO₂ directly beneath the seabed.

This study develops a machine-learning and geospatial-analysis framework to evaluate CO₂ storage site suitability in support of Shell's CCS programme. It employs logistic regression, which is a classical supervised machine learning technique that is relatively straightforward and often regarded as the classification counterpart to linear regression. Although termed 'regression,' the method is slightly misnamed in the context of machine learning, as it is designed to predict categorical outcomes rather than continuous numerical values (Wolfram Research 2025). This study applies logistic regression to estimate the likelihood of a site being suitable for CO₂ storage, using geological features such as shallowest depth, average porosity, and storage permeability, distance to nearest well, distance to licenced polygons, distance to faults and lithostratigraphy as predictors.

2.2 What Makes a Good CO₂ storage site?

A suitable CO₂ storage site must satisfy several geological and engineering criteria. Reservoir quality is central: high porosity (> 15%) ensures sufficient storage capacity, while adequate permeability (> 100 mD) supports efficient injectivity (Bachu 2008, Metz et al. 2005). Capacity must be large enough to store industrially significant volumes, and injectivity sufficient to accept the required injection rates without exceeding fracture pressure. Caprock integrity (seal lithology and thickness) is critical to prevent upward migration, and the depth–pressure–temperature window (typically 800–2500 m) ensures CO₂ remains in the supercritical phase, maximising storage density. Seal geometry and the absence of transmissive faults reduce leakage risks (IEAGHG 2009). Nonetheless, infrastructure proximity (e.g., pipelines, wells, platforms) strongly influences economic feasibility, with re-use of oil and gas infrastructure offering clear advantages for rapid deployment (Ringrose 2020).

2.3 Economic Aspects of CCS

According to the Zero Emissions Platform (ZEP), the cost of CO₂ storage is primarily determined by the location and type of site, along with reservoir size and quality. Onshore sites are generally less expensive than offshore, while depleted oil and gas fields (DOGF) are typically cheaper than deep saline aquifers (SA). Likewise, large reservoirs offer lower costs than smaller ones, and formations with high injectivity are more economical than those with poor injectivity. Given that storage costs can vary by as much as a factor of ten, and that investments in SA carry the risk of identifying reservoirs that prove unsuitable, a risk management framework is required. Such mechanisms would help to realise the significant potential of SA and secure sufficient storage capacity within the necessary timeframe (Global CCS Institute 2011).

This insight can be exemplified by Shell's Quest CCS project in Alberta. Quest captures CO₂ produced during hydrogen generation via steam-methane reforming at the Scotford Upgrader, and transports it via pipeline for injection into a deep saline aquifer approximately 64 km away (IEA Greenhouse Gas R&D Programme (IEAGHG) 2019). The success of this commercial-scale programme underscores how site-specific geological characteristics, such as an onshore location and the use of existing infrastructure, can help reduce costs and demonstrate the importance of risk management frameworks for mitigating storage financial uncertainty. In addition, it illustrates Shell's capacity to integrate capture, transport, and storage operations effectively, offering a practical example of how geological suitability and economic feasibility can be balanced in support of its net-zero objectives.

3 Methodology

3.1 Study Area and Scope

This study is restricted to the analysis of geological sites in the North Sea. For context, however, Table 1 provides a summary of sites where Shell and its joint ventures (JVs) are either already authorised to inject CO₂ or hold the licences and agreements required to obtain such authorisation, along with the remaining consents needed prior to injection.

Table 1: A summary of regulatory instruments and authorisation status relating to Shell-affiliated CO₂ injection and storage activities.

Region / asset	Country	Instrument (authority)	Status
Quest CCS	Canada	AER CO ₂ Disposal Approval 11837C (Alberta Energy Regulator 2024)	Authorised (operating)
Atlas Carbon Storage Hub (for Polaris)	Canada	Carbon Sequestration Agreement (Province of Alberta) (ATCO EnPower and Shell Canada 2024, Reuters 2024)	On path (tenure granted; AER scheme approvals next)
Northern Lights (Aurora, EL001)	Norway	Licence EL001 and storage permit (Norwegian authorities) (Equinor 2024)	Authorised (Phase 1)
Gorgon CCS	Australia	Barrow Island Act 2003 (WA) approvals (Government of Western Australia 2025, 2023)	Authorised (operating)
UKCS storage licences	United Kingdom	Carbon storage licences accepted (NSTA) (North Sea Transition Authority 2023)	Licensed for appraisal (storage permit to follow)

3.2 Data Acquisition and Integration

The dataset used in this study was developed in two stages: the collection of raw geological data and the preparation of a processed dataset for model training. The raw data included geological sequence records from 103 wells in the North Sea aquifer zone, each linked to its geographic position and structural block. These were supplemented with petrophysical measurements such as average shallowest depth, porosity and permeability, together with licensed polygons, pipeline shapefiles, and emissions data.

In addition to reservoir characteristics, lithological seal quality was incorporated into the analysis to capture the trapping potential essential for secure CO₂ storage. The NSTA Collated Well Tops dataset was used to derive caprock thickness as the vertical interval between the top of the sealing unit and the underlying reservoir. For example, in wells where the Bunter Sandstone forms the reservoir and the Bunter Shale provides the seal, the depth to the top of each unit was extracted from the well tops and thickness was calculated as the difference between these two markers (e.g., 3647 m – 3532 m = 115 m) North Sea Transition Authority (NSTA) (2023). These values were then integrated as predictors alongside stratigraphy (categorical), depth, porosity, and permeability (continuous

reservoir properties), as well as distance-based spatial variables (wells, licence polygons, and emissions). This combined approach enabled the logistic regression models to reflect both reservoir quality and sealing capacity within a geospatial framework, thereby ensuring a geologically reliable classification of site suitability.

The dataset used in this study consisted of 56 labelled data points, prepared for supervised machine learning with logistic regression. In total, 35 sites were classified as positives. This group comprised 30 potential storage sites drawn from the CO₂ Stored database, including five located within licensed polygons, one completed storage site, and two sites in commercial operation, the latter reported in Technical Review 2024-TR03 (IEA Greenhouse Gas R&D Programme 2024). The remaining positives were also supported by studies focusing on the North Sea, which identified them as viable candidates for long-term storage. A multicollinearity check was carried out on the CO₂ Stored sites, and one highly correlated feature was removed to reduce the risk of overfitting (SM page 1). The dataset also included 21 negatives, compiled from peer-reviewed studies of the North Sea that determined certain formations and zones to be unsuitable for geological storage. Each site was described by a set of features including stratigraphic grouping, average porosity, storage permeability, and shallowest depth. Geographic coordinates were included in the dataset but not used directly for model training. Instead, they were used to calculate proximity to existing infrastructure and emission zones, as well as to assist in distinguishing potential storage candidates from unsuitable sites. Additional context was provided by geological sequence data from 104 wells in the North Sea aquifer zone, which supplied structural block information, spatial positioning, and petrophysical properties. Additional datasets, such as licensed polygons, pipeline networks, and emissions records, were also integrated. All data were systematically cleaned, harmonised, and prepared prior to model training.

Table 2: Summary of dataset composition used for logistic regression

Category	Count	Details / Source
Positives (total)	35	
CO ₂ Stored (potential sites)	30	Includes 5 within licensed polygons (British Geological Survey 2025)
Completed site	1	(IEAGHG 2024)
Commercial operation	1	(IEAGHG 2024)
Suspended/Planned	1	(IEAGHG 2024)
Potential Sites	2	(SM Page 2)
Negatives (total)	21	From published North Sea studies on unsuitable sites (SM Page 2)
Coordinates (not model features)	–	Used only for proximity calculations and site classification
Total dataset	56	Labelled data points

3.3 Preprocessing and Imputation

Figure 1 shows a Visual comparison of original and imputed values. A fold-wise imputation framework. Although full Multiple Imputation by Chained Equations (MICE) was not feasible given the small sample size, the approach followed the same principle by imputing within k-fold cross-validation (k-fold CV) to avoid information leakage. Both linear regression and mean substitution were applied, with missingness indicators added to retain potential information on data absence. Initially, predictive accuracy varied slightly between runs due to the randomness of fold allocation and model stochasticity; this was stabilised by fixing the random seed (SeedRandom[1234];), which ensured reproducible cross-validation splits. Under this setting, both imputation strategies produced identical predictive accuracy (92.9%), confirming that the findings are robust to the choice of imputation method.

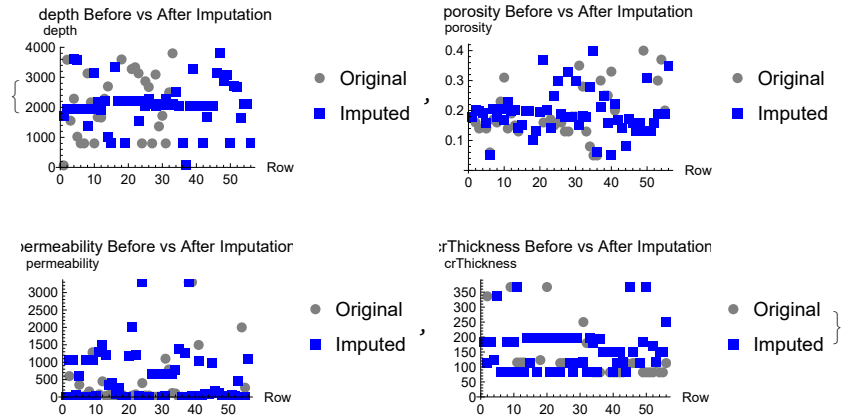


Figure 1: Visual comparison of original and imputed values.

3.4 Firth-penalised Logistic Regression

In order to address potential small-sample bias and the problem of separation in logistic regression, Firth's penalised likelihood approach was employed. Standard logistic regression models the probability of a binary outcome $y_i \in \{0, 1\}$ as

$$\pi_i = \Pr(y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-\mathbf{x}_i^\top \boldsymbol{\theta})},$$

where $\mathbf{x}_i$ denotes the predictor vector (here including an intercept, rescaled depth, and caprock thickness) and $\boldsymbol{\theta}$ the corresponding regression coefficients. The conventional log-likelihood function is

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n [y_i \log(\pi_i) + (1 - y_i) \log(1 - \pi_i)].$$

Firth's method modifies this by introducing a penalisation term based on the determinant of the Fisher information matrix $I(\boldsymbol{\theta})$, defined as

$$I(\boldsymbol{\theta}) = X^\top W X,$$

where $W = \text{diag}\{\pi_i(1 - \pi_i)\}$ is the weight matrix and X the design matrix of predictors. The penalised log-likelihood is then given by

$$\ell^*(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) + \frac{1}{2} \log |I(\boldsymbol{\theta})|,$$

which reduces small-sample bias and ensures finite estimates even under data separation. In practice, this penalised likelihood was maximised numerically using the `FindMaximum` routine in *Wolfram Mathematica*, with the initial parameter vector $\boldsymbol{\theta}_0 = (0, 0, 0)$. The resulting coefficient estimates $\hat{\boldsymbol{\theta}}$ were subsequently used to compute predicted probabilities $\hat{\pi}_i$ for each observation, which provided the basis for model evaluation.

3.5 Code and Data Availability

All geological data and the Mathematica code used for preprocessing, modelling, and visualisation are provided in the supplementary material. A fixed random seed (`SeedRandom[1234];`) was applied during imputation to ensure reproducibility of cross-validation splits. As no seed was enforced during the modelling stage, some results may fluctuate slightly, but remain close to those reported.

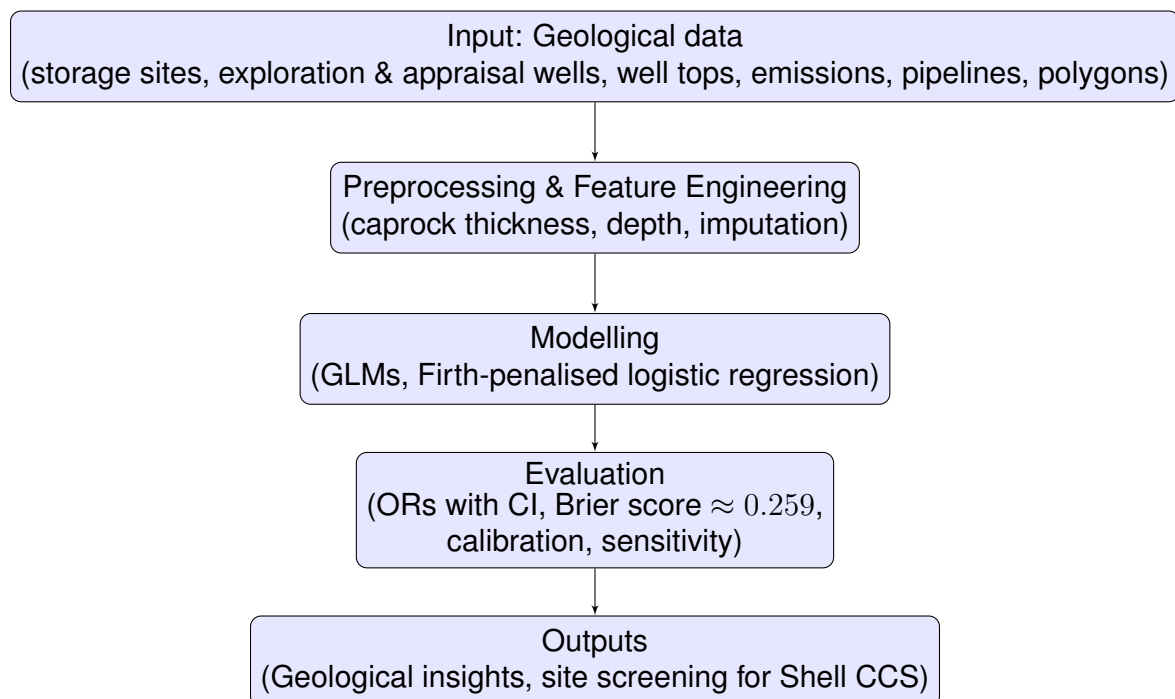


Figure 2: Pipeline of the study, showing progression from geological data inputs through preprocessing and modelling, to evaluation and outputs relevant for Shell's CCS site-screening.

4 Results and Discussion

4.1 Generalized Linear Models

The Generalized linear models in SM (page 3-4) show that Stratigraphy emerges as the dominant and highly significant predictor ($p \ll 0.01$): sandstone-dominated strata strongly increase suitability, with odds multipliers of $\exp(4.7 - 5.6) \approx 100\text{--}270$, whereas shale-dominated strata reduce the odds to about 2% of baseline. Continuous covariates are generally weak, though caprock thickness has a modest positive effect ($\sim 8\%$ higher odds per 100 m), consistent with the expectation that thicker seals confer greater containment, and porosity is borderline positive ($\sim 15\%$ higher odds per percentage point), in line with the view that greater pore space supports injectivity. Depth shows only small, imprecise effects. Overall, the models capture a strong stratigraphic control, with continuous predictors aligning qualitatively with geological principles but offering limited additional explanatory power.

4.2 Error Metrics and Firth-penalised Logistic Regression Performance

Table 3 reports the Firth-penalised logistic regression coefficients ($\hat{\theta}$) with 95% confidence intervals, together with the corresponding odds ratios (OR) and their CIs. For example, a coefficient of -0.10 for predictor 2 implies an odds ratio of 0.90 (95% CI: 0.82 – 0.99), suggesting that higher values of this covariate are associated with reduced suitability, consistent with geological expectations. By contrast, predictor 3 shows a positive estimated effect (OR ≈ 1.22 , 95% CI: 0.72 – 2.06), indicating a tendency towards greater suitability, although the confidence interval overlaps unity.

Table 3: Firth-Penalised Logistic Regression Coefficient Table

Variable	Coefficient	CI Lower (95%)	CI Upper (95%)	Odds Ratio	OR CI Lower	OR CI Upper
Constant	0.5264	−0.0366	1.0895	1.6929	0.9640	2.9727
Porosity	0.4885	−0.3269	1.3039	1.6299	0.7212	3.6838
Depth	0.0130	−0.5646	0.5905	1.0130	0.5686	1.8049
Permeability	−0.2002	−0.9683	0.5680	0.8186	0.3797	1.7646
Seal Thickness	−0.3886	−1.0781	0.3010	0.6780	0.3402	1.3512

Note: No predictors attained statistical significance at the $p < 0.05$ level.

The Brier Score was estimated at 0.259 , indicating performance marginally above chance, where values closer to 0 reflect perfect calibration and 0.25 represents the baseline of a naive classifier with no predictive skill. Whilst the cross-validated Brier Score of 0.259 suggests the model retains modest discriminative ability beyond random assignment, the predicted probabilities remain uncertain and should be interpreted cautiously. The cross-validated accuracy of 54% and Pseudo R^2 of 0.052 further corroborate that the four petrophysical parameters examined porosity, depth, permeability, and seal thickness collectively explain only a limited proportion of the variance in formation suitability. Notwithstanding these constraints, the odds ratio estimates provide directionally consistent findings: porosity (OR = 1.63) exhibited the strongest positive association with suitability, whilst seal thickness (OR = 0.68) and permeability (OR = 0.82) demonstrated negative associations, broadly aligning with established geological principles of reservoir quality and containment integrity. It is important to note, however, that geological heterogeneity including faulting, fracturing, and lithological variability is not fully captured within this probabilistic framework, and the outputs are therefore best regarded as relative screening guidance rather than absolute measures of storage security. For Shell's CCS initiatives, such probabilistic outputs offer valuable early-stage screening tools; whilst not definitive, they highlight relative trends in formation suitability that can inform site ranking and prioritisation. This aligns directly with Shell's net-zero by 2050 commitments, as the integration of penalised regression modelling into site selection workflows may serve to reduce exploration risk, optimise resource allocation, and enhance confidence in long-term carbon storage security across North Sea formations.

4.3 Interpretation Relative to Prior Work

The results from the Firth-penalised logistic regression broadly align with prior CCS studies, notably Sleipner and SACS (Torp & Gale 2004, Eiken et al. 2011), in confirming the directional importance of petrophysical parameters in determining secure CO₂ storage, though without attaining statistical significance in the present study. Porosity emerged as the strongest positive predictor of formation suitability, consistent with its widely recognised role in storage capacity assessments (Dai et al. 2014, Aminu et al. 2017), whilst seal thickness and permeability exhibited negative associations, broadly reflective of the geological trade-off between injectivity and containment integrity. Depth demonstrated a negligible effect, suggesting that at the scale of the present North Sea dataset, overburden alone is insufficient to distinguish suitable from unsuitable formations. The absence of statistical significance across all predictors is most likely attributable to the small and imbalanced dataset ($n = 56$) rather than a genuine absence of geological relationships, a limitation consistent with findings reported in comparable data-constrained probabilistic studies (Nassabeh et al. 2024). The modest cross-validated accuracy and Brier Score further corroborate that the selected petrophysical parameters alone are insufficient to reliably classify formation suitability, underscoring the need for richer and more comprehensive subsurface datasets. This outcome contrasts with the more moderate performances reported in earlier probabilistic and machine learning models (Nassabeh et al. 2024, Vo Thanh et al. 2023, Naghizadeh et al. 2024), highlighting that predictive accuracy is strongly dependent on data quality and volume (Nassabeh et al. 2024). The exclusion of transport cost as a predictor also reflects insights from the Global CCS Institute (Global CCS Institute 2011), which emphasises that economic feasibility is more appropriately addressed through integrated assessment frameworks rather than single cost variables. Notwithstanding these limitations, the directionality of the odds ratio estimates remains geologically consistent, reinforcing established reservoir quality principles whilst demonstrating the potential of Firth-penalised regression as a statistically robust tool for early-stage CCS site screening in data-scarce settings. Incorporating additional geological and geomechanical predictors in future work would be necessary to improve model performance and support more confident site ranking in alignment with net-zero 2050 commitments.

4.4 Implications for Stakeholders

4.4.1 Integration Into Shell's CCS Planning

Whilst none of the predictors attained statistical significance, the directionality of estimates remains geologically consistent, with porosity positively associated with suitability and seal thickness exhibiting a protective effect, broadly aligned with established containment principles (Torp & Gale 2004, Eiken et al. 2011). These findings can be directly inte-

grated into Shell's CCS planning by prioritising formations characterised by higher porosity and favourable seal integrity during early-stage site ranking. For Shell's net-zero by 2050 commitments, incorporating Firth-penalised regression into site selection workflows enables more systematic appraisal of formation suitability, reduces exploration uncertainty, and supports investment decisions aligned with the dual objectives of geological containment security and scalable storage capacity across North Sea formations.

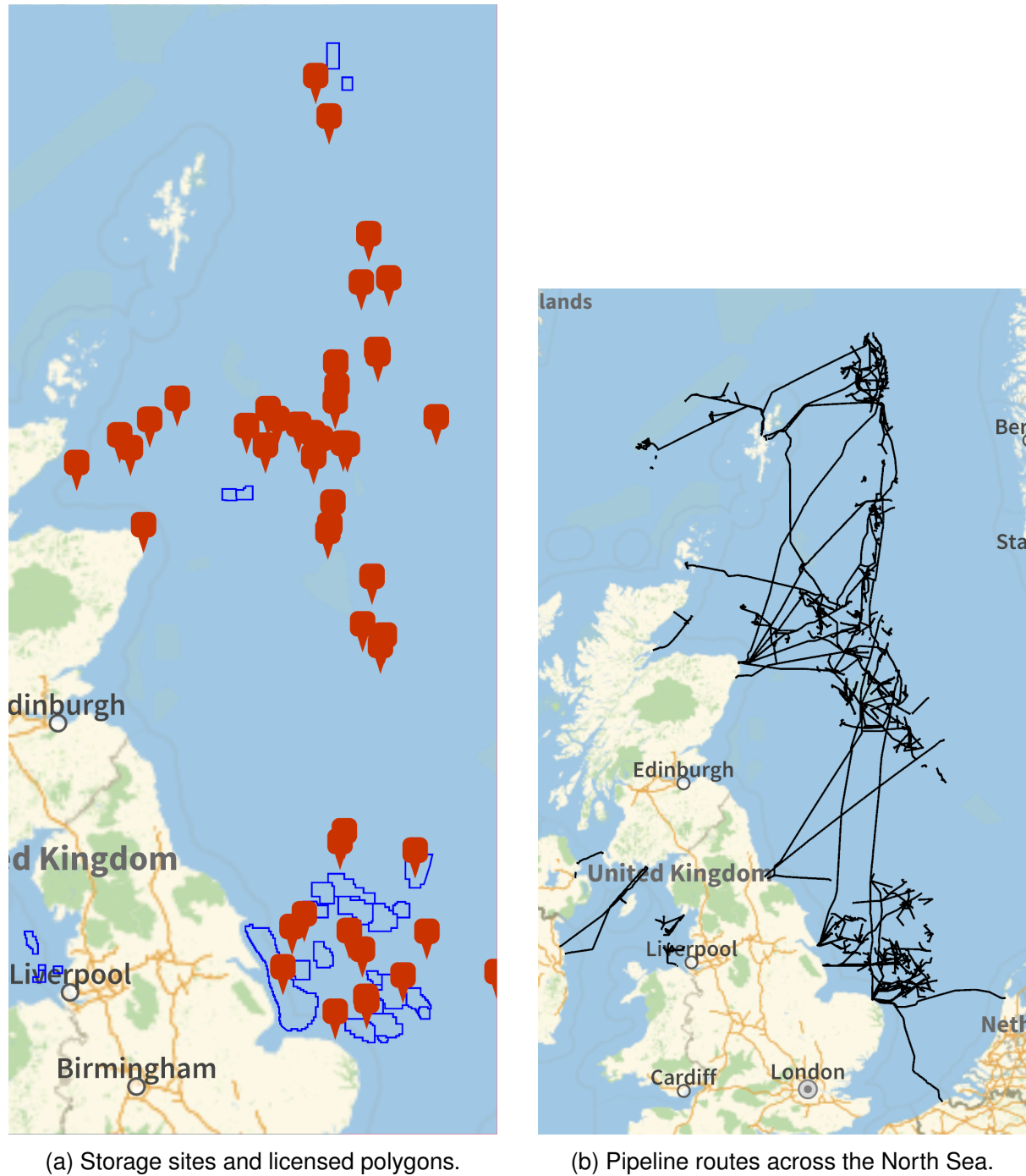


Figure 3: A visual representation of storage site distribution and pipeline routes across the UK North Sea.

4.5 Limitations and Threats to Validity

Despite its contributions, this study is limited by the small dataset ($n = 56$, with only 21 unsuitable formations) and the moderate class imbalance inherent in the available North Sea sample, which may have introduced bias and increased the risk of overfitting, even with the application of Firth penalisation. The restricted sample size further constrained the number of predictors that could be incorporated simultaneously, limiting the scope for examining more complex geological interactions. Missing values were addressed through fold-wise imputation within cross-validation to avoid information leakage, using both linear regression and mean substitution with missingness indicators retained; whilst this ensured reproducibility, the reliance on imputation adds uncertainty, particularly given the limited sample size. Whilst cross-validated accuracy (54%) and Brier Score (0.259) indicate performance marginally above chance, the low Pseudo R^2 (0.052) and non-significant odds ratios across all predictors corroborate that the selected petrophysical parameters alone are insufficient to reliably classify formation suitability at this sample scale. These constraints underline the inherently exploratory nature of the findings and reinforce the need for larger, more balanced, and geologically comprehensive datasets incorporating structural, geomechanical, and economic variables to improve model reliability, predictive calibration, and external validity in support of large-scale CCS site screening.

4.6 Future Work

Future research should focus on expanding the dataset to improve statistical power, balance stratigraphic classes, and reduce reliance on imputation. With larger and more complete data, more flexible modeling approaches could be employed to capture complex interactions between geological predictors. Incorporating economic and infrastructural variables within integrated frameworks would also enhance the practical relevance of the models, aligning technical assessments more closely with decision-making needs for large-scale CCS deployment.

5 Conclusion

This thesis has demonstrated how data science can be applied to the screening of geological formations for CO₂ storage, providing exploratory insights into both the technical and practical aspects of early-stage site assessment. By implementing a Firth-penalised logistic regression model, the study identified porosity as the strongest positive predictor of storage suitability, whilst seal thickness and permeability exhibited negative associations and depth demonstrated a negligible effect. Although porosity, permeability, and depth are widely recognised as critical parameters in the literature (Dai et al. 2014, Aminu et al. 2017), their lack of statistical significance in the present study reflects the constraints of a small and imbalanced North Sea dataset ($n = 56$) rather than a genuine absence of geological relationships. The cross-validated Brier Score (0.259) and accuracy (54%) indicate performance only marginally above chance, meaning that predicted probabilities should be interpreted cautiously and regarded as relative screening guidance rather than absolute measures of storage security. The low Pseudo R^2 (0.052) further confirms that the selected petrophysical parameters alone are insufficient to reliably classify formation suitability at this sample scale. Overall, the directionality of the odds ratio estimates remains geologically consistent, reinforcing established principles that secure CO₂ storage requires favourable reservoir quality and sealing capacity, whilst demonstrating the potential of penalised regression to complement traditional site screening in data-scarce settings. From a practical perspective, such approaches provide Shell and other stakeholders with statistically principled tools for early-stage site ranking, helping to reduce exploration uncertainty and support the expansion of storage capacity needed to achieve net-zero emissions by 2050.

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6 Appendix A. Data Dictionary and Glossary

Variable	Description	Type / Unit
glm	Generalized Linear Model	
depth	Shallowest depth of reservoir below surface	Continuous (m TVDSS)
porosity	Porosity: fraction of reservoir rock volume occupied by pore space (0–1).	Continuous (%)
permeability	Measure of the ease with which fluids can flow through the reservoir rock	Continuous (mD)
stratGroup	Stratigraphy group of reservoir interval (e.g. sandstone-dominated, shale-dominated)	Categorical
crThickness	Caprock thickness (seal lithology)	Continuous (km)
distToLicence	Distance to licence polygon boundary	Continuous (km)
distToPipeline	Distance to nearest pipeline centroid	Continuous (km)
Suitability	Storage suitability label (1 = suitable, 0 = unsuitable)	Binary outcome

6.1 Glossary of Key Terms

CCS Carbon Capture and Storage: capture, transport, and subsurface storage of CO₂.

CCUS Carbon Capture, Utilisation and Storage.

SM Supplementary Material providing additional dataset details and site classifications reference

GLM Generalized Linear Model

$$g(\mu) = \eta = \beta_0 + \sum_j \beta_j x_j,$$

with link function $g(\cdot)$ and an exponential-family likelihood.

Logistic Regression Binary GLM with logit link:

$$p = \Pr(Y = 1 \mid x) = \frac{1}{1 + \exp(-(\beta_0 + \sum_j \beta_j x_j))}.$$

Firth Logistic Bias-reduced logistic regression using Jeffreys prior penalty; mitigates small-sample bias and (quasi-)separation.

OR Odds Ratio for predictor x_j : $\text{OR}_j = \exp(\beta_j)$, interprets multiplicative change in odds per unit x_j .

CI Confidence Interval (e.g. 95%): range expected to contain the true parameter with stated coverage.

Brier Score Mean squared error of predicted probabilities:

$$\text{Brier} = \frac{1}{n} \sum_i (p_i - y_i)^2,$$

where lower values indicate better calibration.

CO₂ Carbon dioxide, a greenhouse gas targeted for capture and geological storage in CCS.